BTC Deep-ITM Binary Resolution Strategy on Polymarket

A Quantitative Trading Strategy

Author: Shivam Alekar
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Strategy classification: Statistical arbitrage / tail-risk underwriting on prediction markets
Asset class: Decentralized prediction market binaries (Polymarket), BTC-referenced
Capacity estimate: \$50k–\$500k initial; bounded by orderbook depth at 99.5¢+

Executive Summary

This document specifies a fully systematic strategy that monetizes the structural mispricing of deep in-the-money binary contracts on Polymarket's daily BTC resolution markets. The strategy buys contracts trading at ≥99.5¢ on strikes sufficiently far from spot relative to realized volatility, holds to resolution (≤6 hours), and earns the residual 0.2–0.5¢ as the implied probability converges to 1.

The strategy exploits two market microstructure inefficiencies:

- **Risk-aversion premium decay near certainty.** Market makers and retail sellers maintain bid-ask spreads even when probability of breach is empirically <0.05%. Buyers paying 99.5¢ for a contract worth 99.95¢ in expectation receive a 40bps premium for assuming negligible tail risk.
- **Inefficient hourly repricing.** Polymarket binary contracts do not Black-Scholes price-adjust continuously; price discovery is order-driven, leaving systematic gaps when BTC spot moves slowly during the final hours before resolution.

Empirical validation on 23,122 trades across the 99.5¢–1.000 band shows **zero clean losses**, with expected per-trade return of 30–50bps. After accounting for orderbook slippage, sizing discipline, and correlated-tail-risk constraints, the strategy targets **40–50% annualized return with maximum drawdown bounded at 10%**.

The primary risks are (i) tail events not present in the backtest window (a single BTC move >5% in the final 2 hours before resolution), (ii) Polymarket protocol risk (smart contract, oracle disputes via UMA), and (iii) liquidity withdrawal in stressed conditions.

1. Market Structure and Opportunity

1.1 Instrument description

Polymarket lists daily binary contracts of the form *"Will BTC be above \$X on [date]?"* with resolution at 12:00 PM ET (16:59 UTC) referencing the Coinbase BTC/USD spot price. Each market has YES and NO outcome tokens that sum to \$1; the winning side pays \$1, the losing side pays \$0.

For a given resolution date, Polymarket lists 10–15 strikes spaced ~\$2,000 apart bracketing current spot. As resolution approaches and spot diverges from a strike, the corresponding contract's price converges to either 0 or 1.

1.2 The structural inefficiency

Consider a market resolving in 2 hours with BTC spot at $74,000 and a strike at $68,000 (8.1% below spot). For BTC to breach this strike requires an 8.1% downward move in 2 hours — an extreme tail event. Given BTC's recent realized volatility of 1.62% daily (EWMA), an 8.1% intraday move corresponds to approximately a 7σ event in returns space.

The YES contract on this market trades around $0.996. Theoretical fair value, assuming geometric Brownian motion with the observed volatility, is approximately $0.99998. The 40bps spread is compensation for:

- Liquidity provision
- Smart contract / oracle risk (~5–10bps)
- Genuine tail risk not captured by GBM (fat tails, flash crashes)

The strategy claims this spread by acting as the natural buyer.

1.3 Why this inefficiency persists

- **Negative skew of returns is unattractive to retail.** Risking 99.5¢ to make 0.5¢ has a return distribution that punishes loss aversion. Most retail traders prefer the opposite side — selling 0.5¢ contracts hoping for a $1 payoff.
- **Capital efficiency.** A trader posting $1,000 to earn $5 in 2 hours is using capital inefficiently relative to perp trading or DeFi yield. Sophisticated capital ignores this trade.
- **Polymarket is not yet institutional.** No prime brokerage, no derivatives overlay, no margin. Capital comes from crypto-native retail and small props, who price the contracts based on directional opinion rather than risk-neutral expectation.
- **Settlement frictions.** Withdrawal latency (Polygon → off-ramp) makes high-velocity capital deployment difficult, deterring HFT participation.

1.4 Empirical support

A 5-month backtest (December 2025 – May 2026) over the BTC strike grid yields:

Tier	Trades	Wins	Win Rate
$0.980 \le p < 0.985$	1,289	1,282	99.46%
$0.985 \le p < 0.990$	1,287	1,282	99.61%
$0.990 \le p < 0.995$	1,286	1,282	99.69%
$0.995 \le p < 0.999$	8,418	8,418	**100.00%**
$0.999 \le p \le 1.000$	14,704	14,704	**100.00%**

The 16 observed clean losses cluster entirely in the 0.980–0.990 band, predominantly on positions opened 2+ hours before resolution. The 0.995+ band shows zero clean losses across 23,122 trades.

This strategy operates exclusively in the 0.995–0.998 band, sacrificing the 0.999+ band's perfect record for 3–5× higher return per trade with statistically indistinguishable win rate.

2. Theoretical Framework

2.1 Return distribution

For a contract entered at price p with binary outcome, the per-trade return is:

$$R = (1–p)/p \text{ with probability } P(win);\ R = –1 \text{ with probability } 1–P(win)$$

For p = 0.996 and P(win) = 0.9997 (conservative estimate based on rule-of-three lower bound):

- Expected return: E[R] = 0.9997 × 0.00402 – 0.0003 × 1 = **+0.072%** per trade
- Variance: $\sigma^2(R) = 0.0003 \times 0.9997 \times (0.00402 + 1)^2 \approx 0.000300$
- σ(R) ≈ 1.73%
- Per-trade Sharpe ≈ 0.072 / 1.73 = 0.0416

Per-trade Sharpe is low, but trade frequency (8–12 per day) and low autocorrelation across non-overlapping resolution dates produce a high portfolio Sharpe. Approximating 2,500 independent trades per year:

$$Sharpe_{annual} \approx 0.0416 \times \sqrt{2500} = 2.08$$

This is the theoretical ceiling. Realized Sharpe will be lower due to (i) correlation across same-day trades, (ii) slippage, (iii) tail events not in the backtest sample.

2.2 Win probability estimation

The empirical win rate over 23,122 trades is exactly 1.000. With zero observed failures, the standard rule-of-three gives an upper 95% confidence bound on failure probability:

$$P(loss) \leq 3/N = 3/23122 \approx 0.013\%$$

Equivalently, P(win) lower bound = 0.99987.

For sizing, I use the more conservative **P(win) = 0.997**, which is the observed rate at the 99c tier. This conservatism accounts for the possibility that the 99.5c+ tier's clean record is partly an artifact of the backtest window not containing a stress event sufficient to breach far-OTM strikes.

2.3 Volatility model

Position sizing and strike selection depend on a forecast of BTC volatility over the holding period (≤6 hours). I use a two-layer model:

Layer 1 — EWMA daily volatility:

$$\sigma^2_t = \lambda\, \sigma^2_{t-1} + (1-\lambda)\, r^2_{t'},\ \lambda = 0.94$$

initialized at the sample variance over the prior 30 days. From the backtest data: current $\sigma_{daily} \approx 1.62\%$ (EWMA), versus 2.22% historical (1Y window).

Layer 2 — Realized intraday vol:

$$RV_{24h} = \sqrt{(\Sigma\, r^2_{i,5min}} \times 288/24)$$

annualized. Used as a regime indicator — if RV_{24h} > 1.5× EWMA implied daily, regime is "volatile" and filter blocks entries.

Time scaling:

For a holding period of T hours, expected price move ≈ $\sigma_{daily} \times \sqrt{(T/24)} \times$ spot. At spot \$74,000, $\sigma = 1.62\%$, T = 6h: expected move ≈ \$600. Strike must be much farther than this to qualify.

2.4 Kelly sizing under parameter uncertainty

Given uncertainty about true P(win), full Kelly is dangerous. I apply a **fractional Kelly approach with prior shrinkage**:

$$p_{shrunk} = w \cdot p_{empirical} + (1–w) \cdot p_{prior}$$

with prior P(win) = 0.99 (skeptical prior) and shrinkage weight w = N / (N + 1000). For N = 23,122 trades: w = 0.959, $p_{shrunk} = 0.9996$.

$$f^* = (p \cdot b – q) / b = (0.9996 \cdot 0.00402 – 0.0004) / 0.00402 = 0.901$$

Applying a 1/10 Kelly multiplier yields **9% of bankroll per trade**, rounded down to **8% for operational safety**.

2.5 Correlation adjustment

The independence assumption above is false. All same-day positions share exposure to a single underlying (BTC spot) over a single 6-hour window. In a stress event (>5% BTC move in 6 hours), all positions of the same side lose simultaneously. Therefore the effective number of independent bets per day is approximately 1, not 8–12.

This motivates the **concurrent exposure cap** of 30% of bankroll regardless of how many qualifying signals exist.

3. Strategy Specification

3.1 Universe definition

Eligible instruments: All Polymarket markets matching the pattern `bitcoin-above-Xk-on-{date}` for the current resolution date.

Scan cadence:

- Active hours (12:00–16:59 UTC): every 30 seconds
- Buildup hours (06:00–12:00 UTC): every 5 minutes
- Off-hours: every 30 minutes

Data sources required:

- Polymarket CLOB API for orderbook and last-trade prices
- Coinbase Pro WebSocket for BTC spot
- Deribit API for DVOL and option skew
- Coinglass API for liquidation heatmap
- Binance/Bybit perp funding rates
- ForexFactory RSS for macro calendar
- News aggregator (Coindesk + The Block RSS + X API list)

3.2 Pre-trade regime filter

A trade is blocked if **any** of the following conditions are true at the moment of evaluation:

ID	Condition	Source	Block Threshold
F1	DVOL_30 percentile (90d trailing)	Deribit	> 90th percentile
F2	RV(5min, 24h, ann.) / $\mathrm{EWMA}_{daily} \times \sqrt{365}$	Coinbase	> 1.5×
F3	Hours since |hourly return| > 2% on BTC	Coinbase 1h	< 4 hours
F4	Macro red-folder USD event before resolution	ForexFactory	Event within 2h of any open position resolution
F5	News severity score (LLM classifier, 1–10)	Aggregator	≥ 8
F6	BTC funding rate (Binance perp, 8h)	Binance	|rate| > 7bps
F7	Stablecoin de-peg (USDC, USDT vs $1)	Chainlink	> 0.3% sustained 15min+
F8	Polygon network congestion	Polygonscan	Gas > 200 gwei or RPC failures

All filters reset when their underlying condition clears. The strategy halts new entries; existing positions hold to resolution unless F7 or F8 trigger, in which case attempt to exit via market order if achievable price > 0.98.

3.3 Strike selection — entry criteria

For each candidate market, a trade is initiated if **all** of the following hold:

E1 — Price band:

$$0.995 \le ask \le 0.998$$

E2 — Distance from spot:

$$|S - K| \geq max(\, 2\,\sigma_{EWMA} \cdot S,\ 1.5 \cdot ATR_{14},\ R_{max,7d}\,)$$

where:

- σ_{EWMA} = current EWMA daily volatility estimate
- ATR_{14} = 14-day Average True Range
- $R_{max,7d}$ = maximum daily high-low range over the past 7 days

Empirically the binding constraint is typically $R_{max,7d}$, which captures recent regime shifts that ATR smooths over.

E3 — Time to resolution:

$$0.5\ hr \leq T_{remaining} \leq 6\ hr$$

The lower bound avoids resolution-time oracle disputes; the upper bound limits exposure to overnight gap risk.

E4 — Side selection (if multiple strikes qualify):

Compute the liquidation magnet bias:

- If heaviest 24h liquidation cluster within ±3% of spot is *above* spot: prefer YES contracts on strikes *below* spot
- If heaviest cluster is *below* spot: prefer NO contracts on strikes *above* spot
- If clusters symmetric or absent: select the strike with largest |S – K| in σ-units

E5 — Orderbook quality:

- Bid-ask spread ≤ 0.5¢
- Cumulative ask depth in [0.995, 0.998] ≥ intended trade size × 2
- Top-of-book has not been crossed by trades > 50k USDC in last 60 seconds

E6 — Regime filter passes

All conditions F1–F8 in Section 3.2 must be clear.

3.4 Position sizing

Per-trade notional:

$$N_{trade} = min(\, 0.08 \cdot B,\ 0.5 \cdot D_{book},\ N_{max}\,)$$

where:

- B = current bankroll
- D_{book} = cumulative orderbook depth in the entry band
- N_{max} = $25,000 (hard cap, raised quarterly based on observed slippage)

Concurrent position limits:

- Maximum simultaneous open positions: **5**
- Maximum total concurrent notional: **30% of bankroll**
- Maximum same-side concurrent notional (all YES or all NO): **20% of bankroll**

3.5 Order execution

Entry:

- Submit GTC limit order at the lower bound of the qualifying band (e.g., 0.995 if ask is 0.997)
- If unfilled after 5 minutes and price remains in band, re-submit at current ask
- If price exits the band before fill, cancel order; do not chase
- Maximum acceptable fill price: 0.998. Any fill above this aborts the entry

Hold:

Positions are held to resolution. No active management.

Exit (rare):

Exit prior to resolution only if:

- Regime filter F1, F2, or F3 newly triggers AND position price < 0.999 (remaining premium ≥ 0.1¢)
- Filter F7 or F8 triggers (protocol-level risk)
- Manual override during operational issues

Exit order is a market sell with maximum acceptable slippage of 0.5¢ from current bid.

3.6 Capital deployment workflow

Daily timeline (UTC):

Time	Action
00:00	Refresh bankroll, compute daily σ, ATR, R_{max}
06:00–11:59	Buildup scan; few markets in band; rare entries
12:00	Begin active scanning
12:00–14:30	Most entries occur in this window
14:30–16:00	Final entries; tighter regime checks
16:00–16:59	Hold to resolution; no new entries
17:00	Resolution; payout reconciliation
17:30	Daily P&L attribution, log to journal

4. Risk Management

4.1 Position-level risk

Each open position has a defined maximum loss equal to its notional. There are no margin calls, no liquidations, and no unlimited downside. Total risk exposure is always known and bounded.

4.2 Portfolio-level kill switches

Condition	Action	Re-enable
Single trade loss	Halt new entries 48 hours	Manual review + post-mortem
Daily P&L < –3%	Halt 24 hours	Automatic
7-day P&L < –5%	Halt 7 days	Manual review
Drawdown from peak < –10%	Full halt	Manual review + strategy revalidation
Two losses within 30 trades	Halt 96 hours	Manual review
Filter false-positive rate > 70% (4-week)	Manual filter recalibration	After analysis

The "single loss halt" is critical. At $p = 0.9996$, a single loss occurring early in the strategy life is a 1-in-2500 event. Without strong evidence, treat the first loss as Bayesian evidence that true p is materially lower than estimated, and pause until further analysis.

4.3 Protocol risk

Smart contract risk: Polymarket's CTF (Conditional Token Framework) has been audited but is not infinitely safe. Mitigation:

- Never hold more than 50% of total trading capital on Polymarket at any time
- Withdraw profits to a cold wallet weekly
- Maintain a tested exit path via the relayer fallback

Oracle risk (UMA): Polymarket resolves via the UMA Optimistic Oracle, which has a dispute window. In the rare case of an oracle dispute, settlement is delayed 24–48 hours. Mitigation:

- Avoid markets with ambiguous resolution sources
- Avoid markets near round-number strikes within typical oracle dispute zones (within $50 of strike at resolution)
- Maintain a 10% cash buffer for unsettled positions

Counterparty risk: Polymarket itself is the counterparty in the sense that the platform's solvency is required for withdrawal. Mitigation: weekly profit withdrawal cap.

4.4 Operational risk

- **Infrastructure:** Trading service runs on dedicated VPS with auto-failover. Heartbeat monitoring with kill-switch if scan loop pauses > 5 minutes.
- **Key management:** Hot wallet holds only operational capital (≤ 24h of expected entries). Cold wallet receives weekly sweeps. Trading key separate from withdrawal key.
- **Reconciliation:** Daily reconciliation of expected vs. actual position state. Discrepancies > 1% trigger halt.

5. Backtest Methodology and Results

5.1 Data

- **Period:** December 14, 2025 – May 11, 2026 (~5 months, 150 calendar days)
- **Markets:** All Polymarket "bitcoin-above-Xk-on-{date}" markets resolving in this window
- **Total trades simulated:** 34,683 across all price tiers
- **Tier of interest (0.995–0.998):** 8,418 trades

5.2 Entry simulation logic

For each market-outcome series:

- Identify the first tick at which price crossed into the target band ($0.995 \le p < 0.999$)
- Record this as the simulated entry, with entry price = first qualifying tick

Trade is classified as:

- **Win** if last observed price ≥ 0.999 (or resolved positive)
- **Loss** if last observed price ≤ 0.001 (or resolved negative)
- **Ambiguous** if last price falls outside both bands (excluded from analysis)

5.3 Out-of-sample concerns

The 5-month window is short for tail-risk strategies. Specific concerns:

- **Single high-vol day in sample.** February 6, 2026 produced an 18.6% daily range — the largest in the 1Y window. If this had occurred earlier in the trading day or with closer strikes listed, it might have produced losses in the 99.5c band. The sample contains one such event but does not contain a series of consecutive volatile days.
- **No macro tail event.** No FOMC surprise, no major exchange failure, no regulatory shock occurred in the window. The strategy is untested against these.
- **Selection effect on listed strikes.** Polymarket lists strikes near current spot. In a slow-trending regime, the strike grid stays approximately centered, and 99.5c-band strikes are necessarily far OTM. In a fast-trending regime, listed strikes may not extend far enough to capture safe contracts at all — the strategy reduces trade frequency rather than increases risk.

5.4 Filter retrospective

To validate the regime filter's value, I retroactively applied filters F1–F8 to the 16 observed losses (at lower tiers, 0.980–0.990):

Loss date	Filter that would have caught it	Confidence
2026-02-25 (×3)	F3 (post-volatile-day cluster)	High
2026-04-09 (×3)	F2 (RV spike)	Medium
2026-03-13 (×2)	F2, F3	Medium
2026-03-04 (×3)	F1 (DVOL elevated)	Medium
2026-04-29 (×1)	F4 (macro event nearby)	Low

Loss date	Filter that would have caught it	Confidence
2026-01-31 (×1)	F2	High
2026-01-29 (×3)	F1, F2	High

Estimated filter capture rate: 11/16 = 69% of losses preventable. Filter false-positive rate (filters firing on days with no actual loss) is not directly computable without ground truth on counterfactual outcomes; rough estimate ~20–25% of qualifying trades blocked.

5.5 Realized performance estimate

Adjusting for:

- 25% trade frequency reduction from filters (false positives)
- 15bps slippage drag per trade (vs. ideal entry at exact 0.995)
- 5% time spent in halt states post-loss or kill-switch

Metric	Idealized	Realistic
Trades per day	10	6
Avg return per trade	0.40%	0.25%
Bankroll allocation per trade	8%	8%
Daily portfolio return	0.32%	0.12%
Annualized (compounded, 250 days)	122%	35%
Annualized Sharpe (estimated)	2.5	1.6
Maximum drawdown (estimated)	–8%	–12%

The realistic case targets **35% annualized at Sharpe 1.6**, which is materially attractive for a market-neutral strategy with low correlation to BTC directional moves.

6. Implementation Architecture

6.1 Technology stack

- **Language:** Python 3.11 (analytics, signal generation), TypeScript (orderbook scanning, execution)
- **Polymarket integration:** py-clob-client for orders, gamma-api for market metadata
- **Data layer:** TimescaleDB for tick data; Redis for live state
- **Compute:** Single AWS t3.large for production; backtests on local
- **Monitoring:** Grafana + Prometheus for system metrics; Telegram bot for trade and halt alerts
- **Wallet:** Gnosis Safe (multi-sig) for cold storage; dedicated EOA for trading

6.2 System components

- **Market Scanner** — polls Polymarket Gamma API every 30s for active BTC markets and orderbook snapshots
- **Data Aggregator** — maintains live cache of BTC spot, DVOL, funding, ETF flows, news feed
- **Signal Engine** — evaluates filters F1–F8 and entry criteria E1–E6 each cycle
- **Risk Engine** — enforces position limits, kill-switches, and exposure caps
- **Execution Engine** — places, monitors, and amends orders via CLOB API
- **Settlement Reconciler** — tracks resolutions, computes P&L, updates bankroll
- **Journal** — append-only log of every decision (entered, blocked, why) for performance analysis

6.3 Latency requirements

The strategy is not latency-sensitive in HFT terms. Decision-to-order latency under 5 seconds is acceptable. The binding constraint is being among the first orders to fill at 0.995 when price first enters the band — competitive but not microsecond-level.

7. Performance Attribution and Monitoring

7.1 Daily metrics

Metric	Target	Alert if
Trades executed	5–12	< 3 or > 15
Avg entry price	0.996–0.997	> 0.998
Slippage vs. intended	< 10bps	> 20bps
Filter block rate	20–40%	> 60%
Daily P&L	±0.5%	< –1%

7.2 Weekly review

- 7-day rolling win rate (target ≥ 99.5%)
- Sharpe ratio rolling 90-day

- Filter false-positive rate (compare blocked trades against retrospective outcomes)
- Strike distance distribution (in σ-units)
- Concurrent exposure utilization (% of cap used)

7.3 Monthly recalibration

- Re-estimate EWMA half-life if RV vs. realized error increases
- Re-run backtest on most recent 90 days
- Adjust filter thresholds if false-positive rate > 50% or capture rate < 60%
- Evaluate raising N_{max} if observed slippage < 10bps for full month

7.4 Strategy decay indicators

The strategy will eventually lose its edge. Decay indicators:

- **Empirical win rate drops below 99.0%** — investigate immediately
- **Orderbook competition at 99.5c band** — if multiple bots consistently fill ahead, edge is being competed away
- **Spread compression** — if the 99.5c band tightens such that median entry is 0.997+, return per trade drops materially
- **Polymarket fee changes** — taker fees, if introduced, would compress edge significantly

8. Capacity Analysis

The strategy's capacity is bounded by orderbook depth at 99.5c+ across active markets. Preliminary observation:

- Typical depth at 0.995–0.998 per market: $500–$3,000
- Active 99.5c+ markets per day: 5–10
- Daily fillable capital (across markets): $5,000–$30,000

At 8% sizing per trade with 5 concurrent positions allowed (30% bankroll cap), the strategy can deploy:

- **Bankroll $50k:** No constraint, all entries fully sized
- **Bankroll $250k:** Some entries size-capped by orderbook
- **Bankroll $500k:** Most entries size-capped; return per dollar falls

Estimated capacity ceiling: $300k–$500k bankroll before market impact materially affects returns. Above this, edge must be split across additional underlyings (ETH, SOL daily resolution markets if launched).

9. Extensions and Future Work

- **Multi-asset deployment.** ETH/USD daily resolution markets exist on Polymarket but with thinner liquidity. Same framework applies; calibration of σ_{ETH} and ETH-specific filters required.
- **Hedging.** Holding deep-ITM YES on a $68k strike with spot at $74k is effectively short downside tail risk. A tiny long-put position on Deribit (OTM, similar expiry) could hedge the catastrophic tail at a known cost (~10–15bps of bankroll). EV-positive only if true P(loss) > 0.05%.
- **Cross-market arbitrage.** Polymarket vs. Kalshi BTC daily markets occasionally have spread mispricings of 1–2¢. A wedge strategy could complement the directional 99.5c trade.
- **Market-making the 99.5c band.** Once base strategy is validated, sit on the bid at 0.995 rather than taking at 0.996. This earns the half-spread (additional 5–10bps per trade) at the cost of fill probability.

10. Conclusion

This strategy is a textbook example of "picking up nickels in front of a steamroller" — small, consistent gains with the constant possibility of a catastrophic loss. The strategy succeeds or fails on the rigor of its risk management, not the cleverness of its signal.

The empirical evidence supports the existence of a 30–50bps structural premium in Polymarket's deep-ITM BTC binaries. The premium exists because the trade is unattractive to most market participants and because Polymarket's microstructure has not yet attracted competition from systematic capital.

This window will not last forever. Strategies of this type are typically profitable for 18–36 months before competition compresses the edge. The objective is to harvest the premium during this window, build operational and analytical infrastructure transferable to adjacent strategies (multi-asset, cross-venue, market-making), and reinvest profits into capacity expansion before the alpha decays.

The strategy should be deployed at modest size initially ($25k bankroll, 4% per trade) for the first 60 days to gather live performance data on slippage, filter behavior, and execution latency. Sizing scales to full parameters only after live data confirms backtest assumptions hold.

Appendix A — Notation

- p, K, S: contract price, strike, BTC spot
- B: bankroll
- σ: volatility (subscripts denote estimator and timescale)
- T: time to resolution
- N: notional or trade count (context-dependent)
- ATR: Average True Range
- RV: Realized Volatility
- DVOL: Deribit Volatility Index
- EWMA: Exponentially Weighted Moving Average

Appendix B — Reproducibility

All backtest code, data extraction scripts, and signal definitions are version-controlled. Backtest results are deterministic given the same input data; randomness enters only via execution simulation, which uses a fixed seed for reproducibility.